Fiscal Year:	FY 2021	**Task Last Updated:**	FY 08/23/2021
PI Name:	Weislogel, Mark M. Ph.D.		
Project Title:	NASA CCF (Capillary Channel Flow) PSI Data Reduction and Benchmark		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
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Human Research Program Elements:	None		
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Space Biology Element:	None		
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PI Email:	mmw@cecs.pdx.edu	**Fax:**	FY 503-725-8255
PI Organization Type:	UNIVERSITY	**Phone:**	503 725-4292
Organization Name:	Portland State University		
PI Address 1:	MCECS Research		
PI Address 2:	P.O. Box 751		
PI Web Page:	https://		
City:	Portland	**State:**	OR
Zip Code:	97207-0751	**Congressional District:**	3
Comments:	Professional affiliations for Dr. Weislogel (updated 1/2/2025): IRPI, LLC 27501 SW 95th Ave, Suite 930 Wilsonville, OR 97070-2905 and Portland State University MCECS Research, P.O. Box 751 Portland, OR 97207-0751		
Project Type:	Ground,Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2015-16 Physical Sciences NNH15ZTT001N-15PSI-C: Use of the NASA Physical Sciences Informatics System – Appendix C
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Contact Monitor:	McQuillen, John	**Contact Phone:**	216-433-2876
Contact Email:	john.b.mcquillen@nasa.gov		
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Task Description:	The Capillary Channel Flow (CCF) EU2-02 experiment conducted aboard the International Space Station (ISS) generated a ~1TB database of over 20,000 test points concerning microgravity bubble transport in an open wedge conduit. Such flows are of interest from a fundamental perspective but are particularly important for engineering applications due to the geometry's ability to passively separate gas and liquid phases for critical spacecraft fluids processing operations. At least six articles connected to the work are listed below. The subject of the experiments has been broadly reported in technical presentations, lectures, and invited seminars. Our single largest work highlighting the breadth of the CCF-EU2 database advertises the still largely untapped depth of the NASA CCF EU2 archive. We repeat that the cumulative CCF EU2-01 tests aboard ISS required 110 24-7 days. Over 200,000 commands were sent to ISS to collect at least 14,344 test points from which over 21 2-phase flow regime maps were created. The value of this 'low-resolution' video data is high as it provides direct measures of passive bubble phase separations in non-circular ducts in the long duration low-g environment of the ISS. Specifically, our research directions for this effort were distilled and prioritized. Due to the immediate application of the research results, numerous applications of the reduced data were and are continuing to be made to systems of open channels with immediate applications to current flight systems design where passive bubble separations are critical such as open channels for liquid sorbent chemistry and bubble management in low-g hydroponic plant watering.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our research is uniquely applicable to low-g environments, many of which are critical to exploration challenges faced by NASA and the commercial aerospace industry. We are presently applying our new knowledge of passive phase separation using conduit geometry to currently pressing problems in fluids management aboard spacecraft. We are intimately aware of the faults and foibles of the state of the art and we are further applying the CCF-EU2 dataset to improving our understanding such that fail-safe capillary two-phase systems may be designed with confidence, greatly reducing risks and the need for crew interaction to maintain system function: fuels and propellants, cooling loops, plant and animal habitats, water processing equipment for life support, medical fluids, food production, etc.
	The data reduction is immediately useful for systems design. The data is useful for developing models to predict flow transitions between regimes determined analytically or numerically. The heuristic transition models are adequate for design, but can be deepened, broadened, and further applied with hopes of designing ubiquitous robust passive no-moving-parts components for troublesome bubble-phase separation plumbing hardware for fuels and life support equipment. We pursued CCF-EU2 archive clean-up activities, the mining of new data, the development of numerical benchmarks, and the applications to low-g passive bubble separating devices. Each will be addressed in order. The CCF-EU2 database on Physical Sciences Informatics (PSI) database is large. The site is expanded to include all 3 CCF-EU2 operations aboard ISS. The data is largely referenced from a single Microsoft Excel spreadsheet with all critical experiment conditions listed and with each associated video hyper-linked. The Excel file employs a variety of macros to sort the data and the most populated maps are provided where one may select any data point in the plots to pull up the associated video. This 'hyper-linking' considerably facilitates the understanding of the general regime behavior and is an enjoyable way to peruse the data. Though valuable for immediate application of the CCF-EU2 results, regime maps of the multiphase flow phenomena in the open wedge channel are only a first step. Mining the data further provides a wealth of results of both fundamental and applied interest. As examples, as only observed in Weislogel et al. (2015) and not studied in depth, it could not be proven whether the bubbles travel along inscribed locations of the channel indefinitely. We observe multi-periodic behavior leading to de/stabilizing beat frequencies, free surface wave interaction that prevents or promotes coalescence outside of expected regime behavior, and local exit geometry that appears to initiate coalescence of otherwise stable bubble train regimes. In the latter part of the CCF-EU2-3 data reduction efforts, several new regimes were identified. The number of such secondary and tertiary phenomena were simply too many to pursue in force during this research. Instead, we pursued only those phenomena that had immediate impacts on parallel applications work. For example, bubble ruptures during separation events in the CCF EU2 channel periodically eject droplets. The accumulated mass of such ejected droplets can be large resulting in large liquid volume losses in the system. The process is uniquely low-g in terms of the large length scales over which it is present. The CCF EU2 data set produced results directly supporting droplet ejection related to wet-lab operations (Turner et al. 2019) and hydroponic plant watering in space (Mungin et al. 2019). Over 90 drop tower experiments were conducted in the connected work (Turner 2020). Far more such phenomena may be identified, none of which has the data been reduced in detail, and each of which could serve as a topic for graduate research in fluid mechanics. We leave such work to ourselves and subsequent parties to pursue from the CCF-EU2 archive. We currently run SE-FIT (2010, 2011) software for capillary statics in complex geometries. For CFD we run commercial code STAR CCM+ (2016) and open source code Geriss (2016). These tools require specific boundary condition data for adequate application to the CCF-EU2 database. This is the special challenge of fluid interface-resolving CFD. However, these data are contained within the CCF-EU2 database. Our CFD can thus be tuned to predict the EU2 data for a selection of the regimes identified experimentally. Unfortunately, despite 4 attempts by 4 students on 4 occasions over 2 years, we were not able to successfully model the flow of inscribed bubbles in such channels. This task remains a challenge. As mentioned, we applied our gleaned knowledge of passive phase separation using conduit geometry to currently pressing problems in fluids management aboard spacecraft. We cite our successes along these lines through which the CCF EU2 database is relevant; i.e., Torres et al. (2020), Mohler and Weislogel (2020), Sargusingh (2016), Graf et al. (2015), Weislogel et al. (2014), Weislogel and Jenson (2013). We are intimately aware of the faults and foibles of the state of the art (i.e., Halvorson (2005), Kossom et al. (1990), Johnson (2002), Puttkamer (2008), Oberg (2008), Clark (1992), Dietrich et al. (2009), Lewis et al. (2006), Weislogel (2017)) and will continue to apply the CCF EU2 dataset to improving our understanding such that failsafe capillary two-phase systems may be designed with confidence greatly reducing risks and the need for crew interaction to maintain system function: fuels and propellants, cooling loops, plant and animal habitats, water processing equipment for life support, medical fluids, food production, etc. References Weislogel, A.P. Wollman, R.M. Jenson, L.M. Sharp, J.S. Geile, J.F. Tucker, B.M. Wiles, A.L. Trattner, C. DeVoe, P.J. Canfield, J. Klatte, M.E. Dreyer (2015) Capillary Channel Flow (CCF) EU2–02 on the International Space Station

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